

Mitigating Physical Inactivity and Prolonged Sitting Among Filipino Desk Workers and Students: An IT-Enabled Workplace Movement Tracking and Nudge System

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I. Introduction

Physical inactivity and prolonged sitting have reached critical levels in the Philippines, driven by the growth of desk-bound office employment, remote work, and digitally dependent academic environments. The Department of Science and Technology – Food and Nutrition Research Institute (DOST-FNRI, 2023) reports that 46% of Filipino adults live a sedentary lifestyle. Furthermore, 75.5% of Filipino adolescents fail to meet recommended daily physical activity standards (Cagas et al., 2022). High screen time and lack of walkable spaces turn screen-based activities into extended periods of physical stillness.

II. Problem Description

The Philippines faces a rising physical inactivity rate, with 46% of adults and up to 75.5% of adolescents failing to meet minimum physical activity standards (DOST-FNRI, 2023). Desk-bound office workers, Remote/BPO employees, and college students are particularly vulnerable to continuous sitting sessions lasting upwards of 8 to 10 hours daily.

While individuals are often aware of the health risks, passive tracking tools (such as standard step counters) fail to enforce structured breaks during long seated periods. Workers and students frequently experience "focus lock," where task overload leads to continuous desk session behavior without mandatory movement intervals. Furthermore, existing health apps do not tailor recommendations to the practical local constraints faced by urban Filipinos, such as limited workspace, indoor settings, or lack of exercise equipment.

III. Proposal Solution

The proposed solution is Galaw-Tech (SmartMove PH), a practical IT-enabled system that actively breaks up continuous sitting periods using standard desktop and mobile hardware. The software runs a lightweight desktop background agent that tracks user activity patterns through keyboard

and mouse usage. When 30 to 45 minutes of continuous typing or sitting is detected, the system triggers a screen-dimming overlay nudge that prompts the user to take a mandatory 2-minute standing micro-break in accordance with physical activity guidelines (DOH, 2023; WHO, 2020). The desktop agent syncs data with a cross-platform mobile app, which displays quick, space-conscious indoor stretching videos (such as seated twists and desk calf raises) tailored for small office cubicles and study desks (OSHC-DOLE, 2022). Users can also enable an optional posture-monitoring module that uses a standard laptop webcam to detect slouching in real time using computer vision.

IV. Conclusion

Physical inactivity caused by continuous sitting is a major health and economic burden in the Philippines that traditional awareness campaigns fail to solve (DOST-FNRI, 2023; UNDP & WHO, 2021). Without active intervention, desk-bound workers and students face severe risks of Non-Communicable Diseases, chronic lower back pain, high out-of-pocket medical costs, and reduced daily productivity (OSHC-DOLE, 2022; PIDS, 2024; PSA, 2024). Galaw-Tech (SmartMove PH) solves this issue by deploying an automated, software-driven system that uses existing desktop and mobile hardware to detect continuous sitting, trigger real-time micro-break nudges, and guide workspace-friendly movement (DOH, 2023; WHO, 2020). By integrating gamification, optional posture monitoring, and institutional analytics, the platform provides a scalable, practical solution to lower sedentary risks, prevent workplace burnout, and build sustainable physical activity habits across Philippine corporate and academic environments.

V. References

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